Name _____ Period _____ Date _____

5-5 Lesson Quiz: Function Operations

short form

1. Let $f(x) = 3x^2 - 2x + 6$ and $g(x) = 7x - 4$. Identify the rule for $g(f(x)) + g(x)$.

Ⓐ $21x^2 - 7x + 34$

Ⓑ $21x^2 - 21x + 50$

Ⓒ $147x^2 - 182x + 62$

Ⓓ $21x^3 - 26x^2 + 57x - 28$

Show your work:

---

4. Let $h(x) = \sqrt{x+3}$ and $k(x) = 2x + 7$. Find the value $\dfrac{h(k(3)) - k(h(1))}{k(h(-2)) - h(k(-3))}$.

Ⓐ $\dfrac{2\sqrt{6} - 2\sqrt{3}}{\sqrt{6} - 7}$

Ⓑ 4

Ⓒ $2\sqrt{6} - \sqrt{12} + 7$

Ⓓ -1

Show your work:

---

5. Let $f(x) = 3x^2 + 2$ and $g(x) = \sqrt{x-4}$. What is the rule for the composition $f \circ g$?

Ⓐ $(f \circ g)(x) = \sqrt{3x^2 - 2}$, domain is all real numbers

Ⓑ $(f \circ g)(x) = \sqrt{3x^2 - 2}$, domain is $x \geq 4$

Ⓒ $(f \circ g)(x) = 3x - 10$, domain is all real numbers

Ⓓ $(f \circ g)(x) = 3x - 10$, domain is $x \geq 4$

Show your work, and explain in one sentence how you found the domain:

Scoring: Question 1 and Question 5 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20.

5-5 Lesson Quiz, short form: answer key

---

1. A, $21x^2 - 7x + 34$**.** $g(f(x)) = 7(3x^2 - 2x + 6) - 4 = 21x^2 - 14x + 42 - 4 = 21x^2 - 14x + 38$; adding $g(x) = 7x - 4$ gives $21x^2 - 7x + 34$. Choice C is $f(g(x)) = 3(7x-4)^2 - 2(7x-4) + 6 = 147x^2 - 182x + 62$ (the wrong order, and $g(x)$ never added); D is $f(x) \cdot g(x) + g(x)$ (a product, not a composition). Savvas quiz item 1, DOK 3, skills F52, F26; taught by Example 5B (substitute the whole inside rule) and Example 6 ($f \circ g$ versus $g \circ f$), with QQD (starred) for the sum. **10 points:** 6 for A, 4 for the work ($7(3x^2 - 2x + 6) - 4$ written and the like terms combined).

4. D, -1**.** Inside values: $k(3) = 13$, $h(1) = 2$, $h(-2) = 1$, $k(-3) = 1$. Then $h(13) = \sqrt{16} = 4$, $k(2) = 11$, $k(1) = 9$, $h(1) = \sqrt{4} = 2$, so the value is $\dfrac{4 - 11}{9 - 2} = \dfrac{-7}{7} = -1$. Savvas quiz item 4, DOK 3, skills F53, F26; taught by working inside out (Example 5A's check line, Example 6's test price, the packet's Exit question). **Bonus 5:** 5 for D with the inside and outside values shown; 3 for D with some of them shown; 2 for D with no work.

5. D, $3x - 10$**, domain** $x \geq 4$**.** $f(g(x)) = 3\left(\sqrt{x - 4}\right)^2 + 2 = 3(x - 4) + 2 = 3x - 10$. The simplified rule is linear, but the inside function $g(x) = \sqrt{x - 4}$ is defined only for $x \geq 4$, so the domain of $f \circ g$ is $x \geq 4$ (choice C is the domain trap). Choices A and B put f inside g: $g(f(x)) = \sqrt{3x^2 + 2 - 4} = \sqrt{3x^2 - 2}$. Savvas quiz item 5, DOK 2, skills F64, F26; taught by Example 5A and the packet's domain checks 1 and 2 (in Example 5A the outside square root sets the limit; here the inside function has the square root, so x must be in the domain of g). **10 points:** 6 for D, 4 for the work (the substitution written and the domain taken from the inside function).

Items left off this form (available for a retake or Form B): Savvas item 2, the revenue function for demand $d(x) = 500 - 2x$ and price $p(x) = x + 5$, $R(x) = 2{,}500 + 490x - 2x^2$ (answer 2,500; 490; 2; DOK 3, skills F53, F26; Example 2, cut in the fast packet); Savvas item 3, the rule and domain of $\frac{f}{g}$ for $f(x) = -3x - 6$ and $g(x) = x^2 - x - 6$ (Part A answer D, $\frac{-3}{x-3}$, DOK 3; Part B answer B and E, 3 and -2 are not in the domain, DOK 2; skills F53, F26; Example 3, cut in the fast packet). The TE quiz has no graphs, so nothing was redrawn.

IXL replacement: SmartScore 70 or above on EKJ, RSP, P9T and QQD replaces this quiz with 20/20.